Dimensionierung

August 24, 2026

[11]:
```python
import numpy as np
```

0.1 Emitterschaltung

Auf was sollen wir U__BE setzen?

[12]:
```python
# gegeben:
I_C=1*1e-3
U_V=15
V_U=20
beta=400
f_ug=100

I_B=I_C/beta
print('I_B =',I_B)
print("\n")

R_C=U_V/(2*I_C*(1+1/V_U))
print(r'R_C =',R_C)
R_C_E12=6.8*1e3
print(r"R_C_Norm =",R_C_E12)
U_C=R_C*I_C
print("U_C [V] =",U_C)
U_C_E12=R_C_E12*I_C
print("U_C_Norm [V] =",U_C_E12)
print("\n")

R_E=R_C/V_U
print("R_E =",R_E)
R_E_E12=330
print("R_E_Norm =",R_E_E12)
U_E=R_E*I_C
print("U_E =",U_E)
U_E_E12=R_E_E12*I_C
print("U_E_Norm =",U_E_E12)
print("\n")

U_CE=U_V-U_E-U_C
```

```
print("U_CE =",U_CE)
U_CE_E12=U_V-U_E_E12-U_C_E12
print("U_CE_Norm =",U_CE_E12)
print("\n")

U_BE=0.6
print("U_BE =",U_BE)
```

```
I_B = 2.5e-06


R_C = 7142.857142857142
R_C_Norm = 6800.0
U_C [V] = 7.142857142857142
U_C_Norm [V] = 6.8


R_E = 357.1428571428571
R_E_Norm = 330
U_E = 0.3571428571428571
U_E_Norm = 0.33


U_CE = 7.5
U_CE_Norm = 7.87


U_BE = 0.6
```

[13]:

```
U_R2=U_BE+I_C*R_E
print("U_R2 =",U_R2)
U_R2_E12=U_BE+I_C*R_E_E12
print("U_R2_Norm =",U_R2_E12)
I_R2=5*I_B
R_2=U_R2/I_R2
print("R_2 =",R_2)
R_2_E12=U_R2_E12/I_R2
print(f"R_2_Norm ={R_2_E12} => R_2_Norm = 68 000")
R_2_E12=68*1e3
print("\n")

I_R1=6*I_B
R_1=(U_V-U_R2)/I_R1
print("R_1 =",R_1)
R_1_E12=(U_V-U_R2_E12)/I_R1
print(f"R_1_Norm ={R_1_E12} => R_1_Norm = 820 000")
R_1_E12=8.2*1e5
```

```python
U_R1=U_V-U_R2
print("U_R1 =",U_R1)
# U_R1=I_R1*R_1  ist derselbe Wert
# print("U_R1 =",U_R1)
U_R1_E12=U_V-U_R2_E12
print("U_R1_Norm =",U_R1_E12)
print("\n")

R_EIN=1/(1/R_1+1/R_2+1/(beta*R_E))
print("R_EIN =",R_EIN)
R_EIN_E12=1/(1/R_1_E12+1/R_2_E12+1/(beta*R_E_E12))
print("R_EIN_Norm",R_EIN_E12)

R_AUS=R_C
print("R_AUS",R_AUS)
R_AUS_E12=R_C_E12
print("R_AUS_Norm",R_AUS_E12)
print("\n")

C_EIN=1/(2*np.pi*R_EIN*f_ug)
print("C_EIN",C_EIN)
C_EIN_E12=1/(2*np.pi*R_EIN_E12*f_ug)
print(f"C_EIN_Norm = {C_EIN_E12} => Norm vor Ort? (100 nF)")
C_EIN_E12=100*1e-9


C_AUS=1/(2*np.pi*R_AUS*f_ug)
print("C_AUS",C_AUS)
C_AUS_E12=1/(2*np.pi*R_AUS_E12*f_ug)
print(f"C_AUS_Norm = {C_AUS_E12} => Norm vor Ort? (1uF)")
C_AUS_E12=1*1e-6
```

```
U_R2 = 0.9571428571428571
U_R2_Norm = 0.9299999999999999
R_2 = 76571.42857142857
R_2_Norm =74399.99999999999 => R_2_Norm = 68 000


R_1 = 936190.4761904761
R_1_Norm =937999.9999999999 => R_1_Norm = 820 000
U_R1 = 14.042857142857143
U_R1_Norm = 14.07


R_EIN = 47330.869808307856
R_EIN_Norm 42551.10535565627
R_AUS 7142.857142857142
R_AUS_Norm 6800.0
```

```
C_EIN 3.3626033862567915e-08
C_EIN_Norm = 3.7403245288606594e-08 => Norm vor Ort? (100 nF)
C_AUS 2.228169203286535e-07
C_AUS_Norm = 2.340513868998461e-07 => Norm vor Ort? (1uF)
```

[14]:
```
f_ug_E12=1/(2*np.pi*R_EIN_E12*C_EIN_E12)
print("f_ug_Norm =",f_ug_E12)
```

```
f_ug_Norm = 37.40324528860659
```

R_1 = 824 kO R_2 = 66.5 kO

0.2 Kollektorschaltung

[15]:
```
I_C=15*1e-3
print('I_E = I_C =',I_C)
I_B=I_C/beta
print('I_B =',I_B)
print("\n")

R_E=U_V/(2*I_C)
print(r'R_E =',R_E)
R_E_E12=4.7*1e2
print(r"R_C_Norm =",R_E_E12)
U_E=R_E*I_C
print("U_E =",U_E)
U_E_E12=R_E_E12*I_C
print("U_E_Norm [V] =",U_E_E12)
print("\n")

U_BE=0.6
print("U_BE =",U_BE)
print("\n")

U_R2=U_V/2+U_BE
print("U_R2 =",U_R2)
print("U_R2_Norm = U_R2 =",U_R2)

I_R2=5*I_B
# print("I_R2",I_R2)
R_2=U_R2/I_R2
print("R_2",R_2)
R_2_E12=39*1e3
print("R_2_Norm =",R_2_E12)
print("\n")
```

```python
R_1=(U_V-U_R2)/(6*I_B)
print("R_1 =",R_1)
R_1_E12=27*1e3
print("R_1_Norm =",R_1_E12)
```

```
I_E = I_C = 0.015
I_B = 3.75e-05


R_E = 500.0
R_C_Norm = 470.0
U_E = 7.5
U_E_Norm [V] = 7.05


U_BE = 0.6


U_R2 = 8.1
U_R2_Norm = U_R2 = 8.1
R_2 43200.0
R_2_Norm = 39000.0


R_1 = 30666.666666666668
R_1_Norm = 27000.0
```

[16]:
```python
R_EIN=1/(1/R_1+1/R_2+1/(beta*R_E))
print("R_EIN =",R_EIN)
R_EIN_E12=1/(1/R_1_E12+1/R_2_E12+1/(beta*R_E_E12))
print("R_EIN_Norm",R_EIN_E12)

R_AUS=R_E
print("R_AUS",R_AUS)
R_AUS_E12=R_E_E12
print("R_AUS_Norm",R_AUS_E12)
print("\n")

C_EIN=1/(2*np.pi*R_EIN*f_ug)
print("C_EIN",C_EIN)
C_EIN_E12=1/(2*np.pi*R_EIN_E12*f_ug)
print(f"C_EIN_Norm = {C_EIN_E12} => Norm vor Ort? verdreifachen: (220 nF)")
C_EIN_E12=220*1e-9


C_AUS=1/(2*np.pi*R_AUS*f_ug)
print("C_AUS",C_AUS)
```

```
C_AUS_E12=1/(2*np.pi*R_AUS_E12*f_ug)
print(f"C_AUS_Norm = {C_AUS_E12} => Norm vor Ort? (10 uF)")
C_AUS_E12=10*1e-6
```

```
R_EIN = 16459.051152928707
R_EIN_Norm 14706.485402273236
R_AUS 500.0
R_AUS_Norm 470.0


C_EIN 9.66975201748343e-08
C_EIN_Norm = 1.0822092344870799e-07 => Norm vor Ort? verdreifachen: (220 nF)
C_AUS 3.183098861837907e-06
C_AUS_Norm = 3.3862753849339433e-06 => Norm vor Ort? (10 uF)
```

[17]:
```
f_ug_E12=1/(2*np.pi*R_EIN_E12*C_EIN_E12)
print("f_ug_Norm =",f_ug_E12)
```

```
f_ug_Norm = 49.19132884032181
```

0.3 PID Regler

[18]:
```
# Widerstände (in Ohm)
R_2 = 1.5 * 1e3
R_3 = 10 * 1e3
R_5 = 1 * 1e3
R_7 = 470 * 1e3
R_8 = 12 * 1e3
R_10 = 100 * 1e3

R_14 = 4.7 * 1e3
R_16 = 100 * 1e3
R_17 = 100 * 1e3
R_18 = 2.2 * 1e3
R_20 = 25 * 1e3
R_23 = 10 * 1e3

R_25 = 10 * 1e3

# Kapazitäten (in Farad)
C_4 = 220 * 1e-9

# Zener-Diode (Z-Spannung in Volt)
U_D1 = 6.8

w_g=10
V_min=1.15
V_max=2.8
```

[19]:
```
# Istwertanpassung: Eingangstiefpass
C_6=1/(w_g*R_17)
print("C_6 =",C_6)
print("\n")

# Istwertanpassung: nicht invertierender Verstärker
R_19=R_18/(V_min-1)
print("R_19 =",R_19)
R_19_E12=1.5*1e4
print("R_19_Norm =",R_19_E12)
print("\n")

# Subtrahierer
R_21=R_22=R_24=R_23
print("R_21=R_22=R_24 =",R_23)
R_21_E12=R_22_E12=R_24_E12=R_23_E12=1*1e4
print("R_21_24_Norm =",R_23_E12)
print("\n")
```

```
C_6 = 1e-06


R_19 = 14666.666666666675
R_19_Norm = 15000.0


R_21=R_22=R_24 = 10000.0
R_21_24_Norm = 10000.0
```

[20]:
```
# P-Regler:
# K_P = 1.5 bis 100
# für P_1 = 0 Ohm und K_P_min=1.5
P_1_min=0
K_P_min=1.5
R_1=(P_1_min+R_2)/K_P_min
print("R_1 =",R_1)
R_1_E12=1*1e3
print("R_1_Norm =",R_1_E12)
print("\n")

# I-Regler:
# K_0= 1 bis 100
P_2_min=0
K_0_min=1
R_6=(K_0_min*R_5)+P_2_min  # was ist mit dem minus?
print("R_6 =",R_6)
```

```python
R_6_E12=1*1e3
print("R_6_Norm =",R_6_E12)
# K_I = 10 bis 1000 1/s
K_I_min=10
C_5=(P_2_min+R_6)/(R_5*R_7*K_I_min)
print("C_5 =",C_5)
print(f"C_5_Norm =  => Norm vor Ort? (220 nF)")
print("\n")

# D-Regler:
# K_D = 0.001 bis 0.1s, K_D=(R_4+P_3)*C_4
K_D_min=0.001
P_3_min=0
R_4=K_D_min/C_4
print("R_4 =",R_4)
R_4_E12=4.7*1e3
print("R_4_Norm =",R_4_E12)
print("\n")

# Inverter V=-1=-R_2/R_1 bei R_2 oberer Widerstand
R_26=R_25
print("R_26 =",R_26)
R_26_E12=1*1e4
print("R_26_Norm =",R_26_E12)

R_9=R_8
print("R_9 =",R_9)
R_9_E12=1.2*1e4
print("R_9_Norm =",R_9_E12)
print("\n")

# Addierer:
R_13=R_12=R_10
print("R_12 = R_13 =",R_10)
print("R_12_Norm = R_13_Norm =",R_10)
R_11=R_10/100
print("R_10 =",R_11)
print("R_10_Norm =",R_11)
```

```
R_1 = 1000.0
R_1_Norm = 1000.0


R_6 = 1000.0
R_6_Norm = 1000.0
C_5 = 2.1276595744680852e-07
C_5_Norm =  => Norm vor Ort? (220 nF)
```

```
R_4 = 4545.454545454545
R_4_Norm = 4700.0


R_26 = 10000.0
R_26_Norm = 10000.0
R_9 = 12000.0
R_9_Norm = 12000.0


R_12 = R_13 = 100000.0
R_12_Norm = R_13_Norm = 100000.0
R_10 = 1000.0
R_10_Norm = 1000.0
```